

DataHow Machine Learning Engineer Coding Challenge

Welcome to the DataHow Machine Learning Engineer coding challenge. Your task is twofold.

1. Develop a predictive model for the final product concentration of a simulated bioprocess for monoclonal antibody (mAb) production, given the provided data in .csv format.
2. Implement the inference microservice for said model given a provided OpenAPI specifications file.

In this challenge, we want to evaluate your capabilities, creativity, and structure in Machine Learning model development, as well as your ability to develop services with clean architecture, following best practices and principles for REST APIs using HTTP.

Under the link [Coding Challenge Interview](#) you will find all necessary data and spec files required to complete this challenge. Please schedule a follow-up interview at your convenience using the scheduling link provided in the email. In this interview, you will have the chance to take us through your code and thought process and discuss your solution to the challenge.

Challenge - Part 1

Develop a model to predict the final titer of a simulated upstream bioprocess for mAb production. The target is the final product titer, a single scalar value per experiment. This dataset is very indicative of the type of data you will work with at DataHow. We will evaluate your data preprocessing, your model evaluation and benchmarking steps, your chosen metrics, and the model architectures you chose. The model's performance is not the main evaluation criterion. We value clean code and clear decisions over squeezing out the last bit of model performance. The data files you will have to use are the following:

- `datahow_interview_train_data.csv`: Training data inputs
- `datahow_interview_train_targets.csv`: Training data targets
- `datahow_interview_test_data.csv`: Test data inputs
- `datahow_interview_test_targets-TEMPLATE.csv`: Placeholder for test data targets

During our interview, we will send you the actual test targets in the same format as `datahow_interview_test_targets-TEMPLATE.csv`. Please prepare your code so you can easily incorporate the final results and discuss them during the interview. Please note that your model's performance may not be optimal, and that's expected. The evaluation

will focus on your approach, code clarity and modularity, and decision making, rather than the model's performance metrics.

Challenge - Part 2

The second part of the coding challenge will include the implementation of an inference server for your model. Next to the data files you will find the OpenAPI specifications file `inference_server_spec.yml` for a REST API exposing the following endpoints:

- GET at /health
- POST at /predict

The payload schema for the POST request is intended as a starting point, not a strict final design. Improvements or alternative implementations are welcome. The response format for the /predict endpoint is intentionally left open for you to define. Bonus points for dockerization, automated testing, linting and formatting checks, and proper serialization and deserialization of data transfer objects (DTOs). Use whatever tools and packages you prefer. We evaluate the implementation by cleanliness of the architecture, deployment procedure of the model, and testing of your microservice.

Additional Notes

- **Time:** There's no strict time limit for this challenge. Take the time you need to deliver high-quality code.
- **Submission:** Publish your project on GitHub or GitLab. Use meaningful commits, demonstrate proper Git practices, and create Pull Requests (PRs) as part of your workflow.
- **README:** Include a README file explaining your architectural choices, the structure of your code, and any decisions made during the development process. Feel free to highlight any challenges you faced and how you solved them.
- **AI Tools:** You are welcome to use AI-assisted tools (e.g. Claude Code, GitHub Copilot). However, during our interview we will ask you questions on your approach, your chosen model, the architectural decisions, as well as any ML and DS theory associated with your solution. Please make sure that you can understand, explain, and defend your decisions instead of pushing the entire responsibility to AI.